1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 11 · 2: Completing the Square

"Completing the square" means putting a quadratic in the form $(x+a)^2+b$

or $a(x+b)^2+c$

The underlying reason we do this is because $\mathbf{x}$ only appears once in the expression (e.g. in $(x+2)^2+3$ vs x^2+4x+7), which makes it algebraically easier to handle. This has a number of consequences:

a. Solving Quadratics

b. Sketching Quadratics

We'll see later that if $y=(x+a)^2+b$, then the minimum point is $(-a,b)$

In Further Maths, completing the square allows us to 'integrate' expressions like:

$\int \frac{1}{x^2-4x+5}dx$

(you will cover integration later this module)

If we have a completed square:

$(x+4)^2-7=0$

we saw at the start of the chapter how we could rearrange to make x the subject.

Indeed using the quadratic formula is actually solving the quadratic by completing the square – it's just someone has done the work for us already!

c. In integration

Chapter 2 · Original slide 12 · Completing the Square Recap

Textbook Note:

The textbook uses the formula

$x^2+bx=\left(x+\frac{b}{2}\right)^2-\left(\frac{b}{2}\right)^2$

and similarly

$ax^2+bx+c=a\left(x+\frac{b}{2a}\right)^2+\left(c-\frac{b^2}{4a}\right)$

My personal judgement is that it's not worth memorising these and you should instead think through the steps. Even the textbook agrees!

Expand:

$(x+\mathbf{9})^2=$
$(x-\mathbf{5})^2=$

What do you notice about the relationship between the bold numbers?

Therefore if we had x^2+12x, how could we write it in the form $(x+a)^2+b$?

Since $(x+6)^2=x^2+12x+36$, we want to discard the 36, so 'throw it away' by subtracting.

Further Examples:

$x^2+8x=$
$x^2-2x=$
$x^2-6x+7=$

Notice that despite the a being negative, we still subtract after the bracket as $(-1)^2$ is positive.

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 13 · Completing the Square

Express $2x^2 + 12x + 7$ in the form $a(x + b)^2 + c$

Factorise out coefficient of x^2. You can leave the constant term outside the bracket.

Complete the square inside the bracket (you should have two sets of brackets)

Expand out outer bracket.

Express $5 - 3x^2 + 6x$ in the form $a - b(x + c)^2$

It may help to write in the form $ax^2 + bx + c$ first.

Chapter 2 · Original slide 14 · Test Your Understanding

Express $3x^2 - 18x + 4$ in the form $a(x + b)^2 + c$

Express $20x - 5x^2 + 3$ in the form $a - b(x + c)^2$

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 15 · Solving by Completing the Square

Solve the equation:

$3x^2 - 18x + 4 = 0$

Reflections: Previously we factorised out the 3. This is because $3x^2 - 18x + 4$ on its own is an expression, so dividing by 3 (instead of factorising) would change the expression.

However, in an equation, we can divide both sides by 3 without affecting the solutions.

Chapter 2 · Original slide 16 · Exercise 2C/2D Pure Y1 Page 23-24

1 Complete the square for the expressions:

a $x^2 + 4x$
b $x^2 - 6x$
c $x^2 - 16x$
d $x^2 + x$
e $x^2 - 14$

2 Complete the square for the expressions:

a $2x^2 + 16x$
b $3x^2 - 24x$
c $5x^2 + 20x$
d $2x^2 - 5x$
e $8x - 2x^2$

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 17 · Exercise 2C/2D Pure Y1 Page 23-24

4 Given that $x^2 + 3x + 6 = (x + a)^2 + b$ find the values of the constants a and b

5 Write $2 + 0.8\text{x} - 0.04x^2$ in the form $A - B(x + C)^2$ where $A \text{and} B$are constants to be determined

4 By completing the square, show that the solutions to the equation $x^2 + 2bx + c = 0$ are given by the formula $x = -b \pm \sqrt{b^2 - c}$. (4 marks)

Chapter 2 · Original slide 18 · Proving the Quadratic Formula

If $ax^2 + bx + c = 0$, prove that $x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Just use exactly the same method as you usually would!

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 26 · 4: Quadratic Graphs

Recall that x refers to the input of a function, and the expression $f(x)$ refers to the output. For graph sketches, we often write $y = f(x)$, i.e. we set the y values to be the output of the function.

y

A 'sketch' in maths has a clearly defined meaning:

• We care about the general shape of the graph, not exact points.

• The axes should have no scale.

• Generally the only coordinates indicated are intercepts with the axis or other points of interest (e.g. intersections of multiple graphs)

Features needed in sketch of quadratic?

Recall a root of a function is where the output, in this case the y value, is 0.

Roots

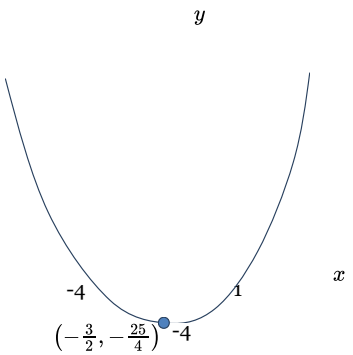
x

General shape: Smiley face or hill?

y-intercept

Turning point

Chapter 2 · Original slide 27 · Example



Sketch the graph of $y = x^2 + 3x - 4$ and find the coordinates of the turning point.

Roots:

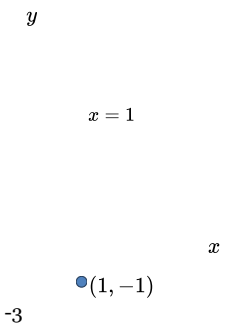
y-intercept:

Turning point:

Recall that if $f(x) = (x + a)^2 + b$, the minimum output is b and $-a$ is the x value which minimises it. i.e. Turning point is $(-a, b)$

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 28 · Example



Sketch the graph of $y = 4x - 2x^2 - 3$ and find the coordinates of the turning point. Write down the equation of the line of symmetry.

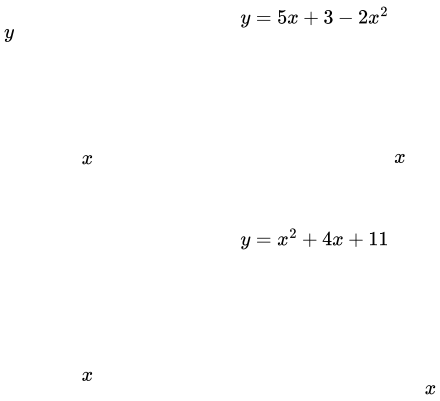
Roots:

y-intercept:

Turning point:

Line of reflection:

Chapter 2 · Original slide 29 · Test Your Understanding



Sketch the following, indicating any intercepts with the axis, the turning point and the equation of the line of symmetry.

a c

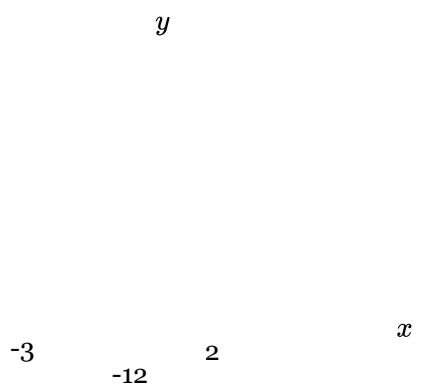
$y = x^2 + 4$

b d

$y = x^2 - 7x + 10$

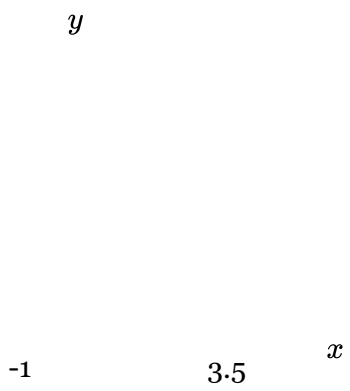
1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 30 · Determining the Equation using a Graph



Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$.

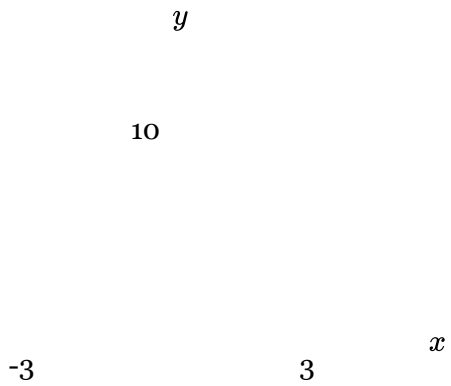
Chapter 2 · Original slide 31 · Further Example



Determine the equation of this quadratic graph, in the form $y = ax^2 + bx + c$, where a, b, c are integers.

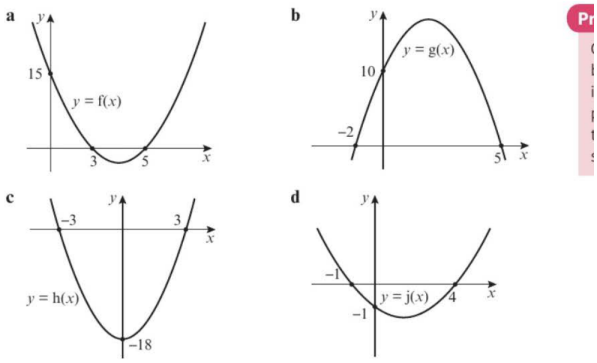
1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 32 · Test Your Understanding



Determine an equation of this quadratic graph.

Chapter 2 · Original slide 33 · Exercise 2F Pure Y1 Page 30



2 These sketches are graphs of quadratic functions of the form $ax^2 + bx + c$. Find the values of a,b,c for each function.

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 34 · Exercise 2F Pure Y1 Page 30

3 The graph of $y = ax^2 + bx + c$ has a minimum at (5,-3) and passes through (4,0). Find the values of a,b,c.

[MAT 2003 1H] Into how many regions is the plane divided when the following three parabolas are drawn?

$y = x^2$
$y = x^2 - 2x$
$y = x^2 + 2x + 2$

x

Extension Question:

y

Chapter 2 · Original slide 35 · Starter

How many distinct real solutions do each of the following have?

$x^2 - 12x + 36 = 0$
$x^2 + x + 3 = 0$
$x^2 - 2x - 1 = 0$

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 36 · 5: The Discriminant

$ax^2 + bx + c = 0$

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Note: Roots of a function f are the values of x such that $f(x) = 0$.

Similarly the roots of an equation are solutions to an equation in the form $f(x) = 0$

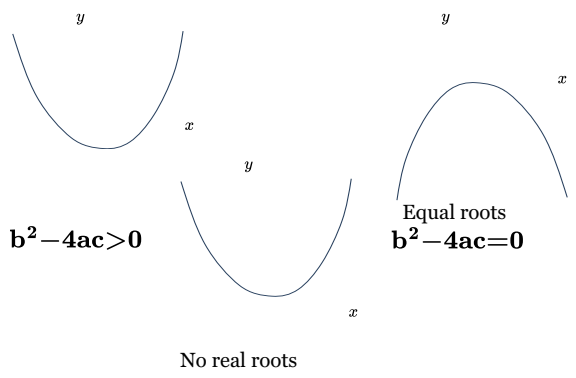
Looking at this formula, when in general do you think we have:

- No real roots?
- Equal roots?
- Two distinct roots?

$b2 - 4ac$is known as the discriminant.

Because adding 0 or subtracting 0 in the quadratic formula gives the same value.

Chapter 2 · Original slide 37 · 5: The Discriminant



We say the graph “touches” the axis.

Distinct real roots

$b^2-4ac<0$

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 38 · Just for your interest…

Why do we say “Equal Roots” not “One root”?

It is due to the Fundamental Theorem of Algebra:

“Every polynomial of order n has exactly n roots.”

$x^2 - 12x + 36 = 0$

Using the quadratic formula gives us the same value in both $+$ and $-$ cases: $x = 6$.

You might wonder why we say “*it has one repeated root*” or “*it has equal roots*”, i.e. indicating we have 2 roots (but with the same value). Why not say it has 1 root?

Despite the theorem being a simple statement, it was only until 1806 that it was first proven by Argand. Clearly by using the quadratic formula we can show a quadratic equation has 2 roots. We can use similar formulas to show that a cubic has 3 roots and a quartic 4 roots. But there is provably no such formulae for order 5 (quintics) and beyond. So we have to prove for example that 5 roots exist for a quintic, despite us having no way to find these exact roots!

One side result of the Fundamental Theorem of Algebra is that every polynomial can be written as a product of linear and/or quadratic expressions.

Leibniz claimed in 1702 that a polynomial of the form $x^4 + a^4$ cannot be written in this way. He then got completely burned by Euler in 1742 who managed to do so:

$$x^4 + a^4 = \left(x^2 + a\sqrt{2}x + a^2\right)\left(x^2 - a\sqrt{2}x + a^2\right)$$

A polynomial is an expression with non-negative integer powers of x, i.e.

$a + bx + cx^2 + dx^3 + \ldots$

All linear, quadratic and cubic expressions are examples of polynomials.

The order of a polynomial is its highest power of x. So the order of a quadratic is 2, and a cubic 3.

These roots might be repeated or might not be ‘real’ roots. $\sqrt{-1}$ is known as a complex number, which you will encounter if you do FM. But it is still a value!

The theorem means that a quadratic (order 2) will always have 2 roots. This is why you should say “no real roots” when $b^2 - 4ac < 0$ rather than “no roots”, because there are still roots – it’s just they’re not ‘real’! Similarly we must say “*equal roots*” because there are still 2 roots.

There are various other ‘Fundamental Laws’. The ‘FL of Arithmetic’ you encountered at KS3, which states that “*every positive integer* > 1 *can be written as a product of primes in one way only*”. You will encounter the ‘FL of Calculus’ in Chapter 13.

Chapter 2 · Original slide 39 · Quickfire Questions

Equation	Number of Distinct Real Roots	Discriminant
$x^2 - 4x + 1 = 0$		
$x^2 - 4x + 4 = 0$		
$2x^2 - 6x - 3 = 0$		
$x - 4 - 3x^2 = 0$		
$1 - x^2 = 0$		
$x^2 + 3x + 4 = 0$		

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 40 · Problems involving the discriminant

8. The equation $x^2 + 2px + (3p + 4) = 0$, where p is a positive constant, has equal roots.

(a) Find the value of p. (4)

(b) For this value of p, solve the equation $x^2 + 2px + (3p + 4) = 0$. (2)

a)

Tip: Always start by writing out a, b and c explicitly.

b)

Chapter 2 · Original slide 41 · Exercise 2G Pure Y1 Page 32

2 Find the values of k for which $x^2 + 6x + k = 0$ has two real solutions.

3 Find the value of t for which $2x^2 - 3x + t = 0$ has exactly one solution.

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 42 · Exercise 2G Pure Y1 Page 32

4 Given that the function $\mathrm{f}(x) = sx^2 + 8x + s$ has equal roots, find the value of the positive constant s.

5 Find the range of values of k for which $3x^2 - 4x + k = 0$ has no real solutions.

Chapter 2 · Original slide 43 · Exercise 2G Pure Y1 Page 32

7 $\mathrm{h}(x) = 2x^2 + (k + 4)x + k$, where k is a real constant.

a Find the discriminant of $\mathrm{h}(x)$ in terms of k.

b Hence or otherwise, prove that $\mathrm{h}(x)$ has two distinct real roots for all values of k.

6 The function $\mathrm{g}(x) = x^2 + 3px + (14p - 3)$, where p is an integer, has two equal roots.

a Find the value of p.

b For this value of p, solve the equation $x^2 + 3px + (14p - 3) = 0$.

1.09 Quadratics: Problem Solving

Chapter 2 · Original slide 44 · Test Your Understanding

$x^2 + 5kx + (10k + 5) = 0$

where k is a constant.

Given that this equation has equal roots, determine the value of k.

Find the range of values of k for which $x^2 + 6x + k = 0$ has two distinct real solutions.

Chapter 2 · Original slide 45 · Exercise 2G Pure Y1 Page 32

Extension Questions:

1

[MAT 2009 1C] Given a real constant c, the equation

$x^4 = (x - c)^2$

Has four real solutions (including possible repeated roots) for:

A) $c \leq \frac{1}{4}$

B) $-\frac{1}{4} \leq c \leq \frac{1}{4}$

C) $c \leq -\frac{1}{4}$

D) all values of c

2

[MAT 2006 1B] The equation $\left(2 + x - x^2\right)^2 = 16$ has how many real root(s)?

3

[MAT 2011 1B] A rectangle has perimeter P and area A. The values P and A must satisfy:

A) $P^3 > A$

B) $A^2 > 2P + 1$

C) $P^2 \geq 16A$

D) $PA > A + P$